Reconciled Standard Model table — fiber-layer ↔ bulk-Shilov dialects in one tier-honest object (v0.1)

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Reconciled SM table v0.1

How to read it

BST has two geometric dialects for the same particles, and this table puts them side by side so the Periodic Table can be built on one consistent object:

- Fiber-layer (April catalog; `bst_particles.json` / `bst_forces.json`): charge = S^1 winding number; color = Z_3 circuit on CP^2; weak = Hopf fibration $S^3 \to S^2$.
- Bulk-Shilov (May A1 / Hall algebra): leptons = Shilov-boundary K-types; quarks = bulk K-types; charge = SO(2) weight; generation = winding-mode coordinate W_n.

They agree on the integers and the qualitative story; they are not yet merged (that's the Hall-engine program). Two flags per cell: - Tier = D (mechanism derived) / I (matches <1%, mechanism partial) / C (conditional/structural-zero) / S (>2% or qualitative). - Constr. = DERIVED (substrate construction proved) or ASSIGNED (hand-assigned label; canonical basis #402 would derive it).

A. Leptons (Shilov-boundary sector)

Particle	Fiber-layer winding	Bulk-Shilov / level	Charge (mechanism)	Color	Spin	Mass (BST)	Tier	Constr.
e^-	1 full S^1 turn on Shilov $S^4 \times S^1$, n=−1	Shilov K-type, k=1 (below Wallach k_min)	$-1 = S^1$ winding no. (= SO(2) weight)	none	½	anchor (m_e)	— (unit)	DERIVED (simplest persistent winding)
μ^-	e^- circuit threaded through D_IV^3	gen-2 Shilov K-type, shares W_1	−1	none	½	$(24/\pi^2)^6 = 207$ (also 9·23)	I (0.004%)	ASSIGNED
τ^-	e^- circuit at full D_IV^5 depth	gen-3 Shilov K-type, shares W_2	−1	none	½	49·71 = 3479	I (0.05%)	ASSIGNED

Particle	Fiber-layer winding	Bulk-Shilov / level	Charge (mechanism)	Color	Spin	Mass (BST)	Tier	Constr.
ν_e	Z_3 Goldstone / vacuum ground mode, n=0	D_IV^5 vacuum mode	0 (zero winding)	none	½	0 exactly	C (structural-0)	DERIVED (vacuum mode → 0)
ν_μ	1st vacuum excitation	vacuum mode 2	0	none	½	(7/12)α^2·m_e^2/m_p	I	ASSIGNED
ν_τ	2nd vacuum excitation	vacuum mode 3	0	none	½	(10/3)α^2·m_e^2/m_p	I (~1%)	ASSIGNED

Neutrino character: Dirac (B−L conserved, no 0$\nu\beta\beta$) — resolved this session, per BST_NeutrinolessDoubleBeta.

B. Quarks (bulk sector — confined)

Particle	Fiber-layer winding	Bulk-Shilov	Charge (mechanism)	Color	Spin	Mass (BST)	Tier	Constr.
u	⅔ S^1 + ⅓ Z_3 on CP^2	bulk K-type, shares W_0	+⅔ forced by Z_3 closure	yes	½	c_3·N_c^2/(rank^2·g)·m_e	I (1.4%)	charge DERIVED / mass ASSIGNED
d	−⅓ S^1 + ⅓ Z_3	bulk K-type, W_0	−⅓ forced by Z_3 closure	yes	½	c_3·(N_c^3−rank^3)/(rank^2·g)·m_e	I (1.5%)	charge DERIVED / mass ASSIGNED
c	u at D_IV^3 depth	bulk, W_1	+⅔	yes	½	m_s·c_3 (cascade)	S (9–12%)	ASSIGNED
s	d at D_IV^3 depth	bulk, W_1	−⅓	yes	½	m_d·19 (Ogg cascade)	S (9%)	ASSIGNED
t	u at D_IV^5 depth	bulk, W_2	+⅔	yes	½	(1−α)·v/$\sqrt{2}$	I (0.8%)	ASSIGNED
b	d at D_IV^5 depth	bulk, W_2	−⅓	yes	½	m_t/(C_2·g) = m_t/42	I (0.8%)	ASSIGNED

Confinement (DERIVED, FRAMEWORK-PLUS): a single quark is a ⅓ Z_3 circuit — an *open* circuit with no Shilov boundary value; only color-singlet composites (3 quarks, or q-q̄) close and project. Fractional charge is forced by the requirement that the three S^1 partial-windings sum to an integer (Z_3 closure ? integer total charge ? confinement). This is the cleanest DERIVED mechanism in the table.

C. Bosons / force carriers (the "couplings" block)

Boson	Fiber-layer	Bulk-Shilov role	Mediates	Mass	Tier	Constr.
γ	S^1 phase oscillation, zero winding	S^1 phase channel	EM	0	D (zero topological cost)	DERIVED
gluon ×8	Z_3 partial winding on CP^2, adjoint	color channel	color	0	D (count 8 = N_c^2−1 exact)	DERIVED (count)
W±	charged Hopf $S^3 \to S^2$ excitation	flavor/generation change (chiral)	weak CC	n_C·m_p/(8α) ≈ 80.36 GeV	I (m_W/m_Z 0.05%)	ASSIGNED
Z^0	neutral $S^1 \oplus$Hopf mix	neutral current	weak NC	m_W/cos θ_W	I	ASSIGNED
H	Kähler radial vacuum mode	bulk Kähler radial	mass generation	rank·g/N_c^2·m_W ≈ 125 GeV	I (0.05%)	ASSIGNED
graviton	does not exist	—	gravity = S^1 integrability/boundary condition (NOT a force)	—	D (absence predicted)	DERIVED

D. Composites (the "constructed-from" layer)

Object	Construction	Bulk level	Charge	Mass (BST)	Tier	Constr.
proton	3 quarks, Z_3 closure on CP^2/T^2	bulk k=6, C_2=6 (fundamental)	+1	$6\pi^5$·m_e	D (0.01%)	DERIVED (= YM mass gap)
neutron	proton's Hopf sibling (udd)	k=6 + Hopf correction	0	(m_n−m_p)/m_e = 91/36 ≈ n_C/rank	I	DERIVED (quasi-eigentone, unstable)
π, K, hadrons	q-q̄ / 3q closures	bulk resonances	—	(Paper 108 §2.9, 16 entries)	I	ASSIGNED

E. The charge↔color link, in one line

Electric charge = S^1 winding number (= SO(2) weight); color = Z_3 closure on CP^2; the fractional quark charge is forced by the color closure (3 quarks must sum to integer S^1 winding). Color closure ▯ integer total charge ▯ confinement. This is one DERIVED mechanism, not three facts.

F. Honest scorecard (the gap map this table makes visible)

- DERIVED cells: proton mass / YM gap (D, 0.01%), α^{-1}=137, gluon count 8=N_c^2−1, photon & gluon masslessness, fractional-charge-from-Z_3-closure, graviton absence, the scheme-invariant mixing spine ($\sin^2\theta$_W=rank/N_c^2, PMNS /N_max), ν masslessness.

- ASSIGNED (the bulk of the masses): μ, τ, all six quark masses, W, Z, H, the heavier ν — formula matches, substrate construction hand-assigned. These flip to DERIVED only when the canonical basis (#402) lands — that is the single biggest lever in the program.
- S-tier rough: the gen-2 quark cascade (s, c) at 9–12%.
- Two-dialect residual: the fiber-layer ↔ bulk-Shilov columns are not yet a single map; merging them is the Hall-engine + canonical-basis work.

G. Tier-gate on the program's load-bearing rep-theory (Keeper verification)

Pre-grounding E0/E1 (#400/#401) so the bet's *mathematical* premises are checked separately from its *physics* identifications:

- VERIFIED standard math: B_2 has 4 positive roots (#positive roots of B_n = n^2; $B_2 \rightarrow 4$). By Gabriel's theorem (ADE) extended to valued quivers/species by Dlab–Ringel (B,C,F,G), a finite-type species has #indecomposables = #positive roots. So the finite B_2 species over GF(2) has exactly 4 indecomposables — Elie's premise is sound, and 4 is indeed too few for the SM's particle content.
- VERIFIED standard math: the affine $\hat{B}_2$ is tame type → indecomposables fall into tube families (a P^1-worth), giving an infinite, structured spectrum — the room a generation tower + mass tower would need.
- STILL THE BET (must be tested, not assumed): that extensions = SM interaction vertices and the 3 tubes = the 3 generations. The representation theory is standard; the *physics dictionary* onto it is the hypothesis E0 + the 3-tubes check are designed to confirm or break. Keep this line bright: do not let "3 tubes = 3 generations" be stated as derived until E1 closes it.

— Keeper, 2026-05-28 v0.1. This is the consistent backbone for the Periodic Table (#403): same particles, both dialects, every cell flagged DERIVED-vs-ASSIGNED + D/I/C/S. The ASSIGNED→DERIVED conversion is the canonical-basis lever; the proton (D, 0.01%) is the anchor; the mixing spine is the rigorous edge; the quark cascade is the soft spot. Next: hand to Grace as the Periodic-Table coordinate fill, and re-flag cells DERIVED as the engine lands.